

CLEARSY

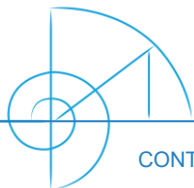
Safety Solutions Designer

AIX
LYON
PARIS
STRASBOURG

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July 2026

Hackathon 2026



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About CLEARSY

- ▶ Experts in formal methods for safety-critical systems
- ▶ More than 20 years of experience with the B Method
- ▶ Development of tools such as Atelier B, ProB integrated tools and related technologies
- ▶ Working with railway, transportation and other high-integrity systems

The B Method

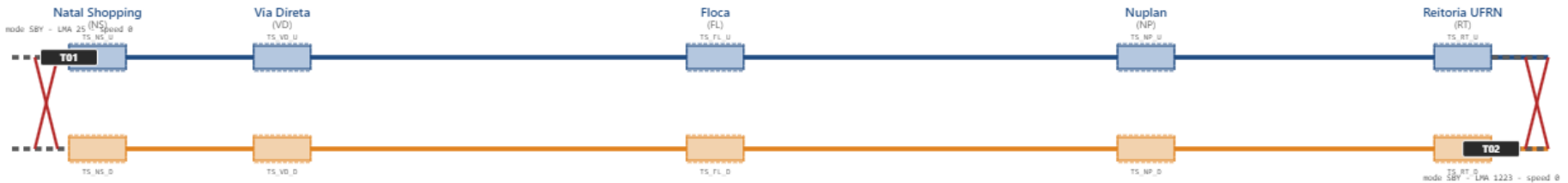
- ▶ The B Method has been successfully used in the railway industry since the development of the automation system for **Paris Metro Line 14**, one of the world's first fully automated metro lines.
- ▶ Today, formal methods are used in many railway applications, including:
 - ▷ Platform screen doors
 - ▷ Automatic train protection
 - ▷ Driverless train control
 - ▷ Signalling and safety systems

The Challenge – Line 0



Your Mission

- ▶ During this hackathon, you will specify the control system of an autonomous metro line using the B Method.
- ▶ Rather than working on an existing railway, we created a fictional one: **Line 0**, connecting Natal Shopping to the UFRN Rectorate



Hackathon Schedule

- ▶ **Duration: 5 days**
- ▶ **Daily session:**
 - ▶ 🕒 **14:50 – 16:30** (except on the last day, when the time will be reduced for result calculation purposes)
- ▶ Each day introduces new concepts and extends the specification developed during the previous sessions.

Daily Material

- ▶ Each session includes:

- ▷  An incomplete B machine to be completed
- ▷  A short presentation introducing the day's topic
- ▷  A description of the activity and objectives

- ▶ The first day also includes a map of the metro line showing all the landmarks.
- ▶ Your goal is to progressively build the complete specification throughout the week.

Systems to be developed:

- ▶ Day 1 – **Database** : a shared database used by all the systems.
- ▶ Day 2 – **CBI** (Computer-Based Interlocking) : The route manager. Locks sections, opens signals.
- ▶ Day 3 – **ZC** (Zone Controller) : The radio dispatcher on the wayside. Computes each train's Movement Authority and sends it by radio.
- ▶ Day 4 – **Onboard ATP** (Automatic Train Protection) : The safety brain inside the train. Receives the LMA, supervises the speed, applies the emergency brake.
- ▶ Day 5 – **ATS** (Automatic Train Supervision) : The control room. Holds the timetable. Decides *when* to send a new dispatch.

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Recommendation: Use Atelier B to specify and fix errors, but test your machines in ProB before submitting them. The automatic analysis used for scoring will be performed with ProB.

Daily material is available at
<https://github.com/dalayalmeida/Hackathon2026>

Results must be submitted as a **.zip** file named after the participant and sent to [**dalay.almeida@clearsy.com**](mailto:dalay.almeida@clearsy.com).

